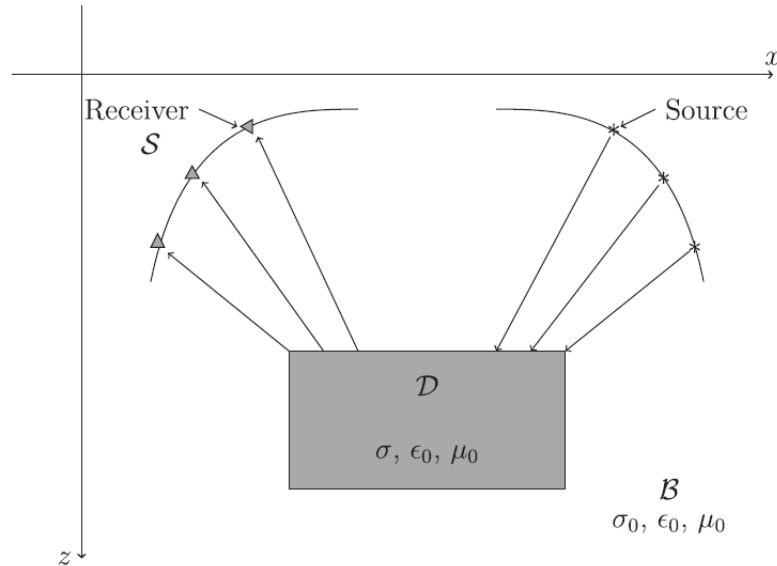


Contrast-source inversion

Ketil Hokstad, March 2016

Wave equations



Wave equation on $\mathcal{B}$ (background):

$$\partial_i \partial_i G_0(\mathbf{x}, \mathbf{x}') + \kappa_0^2 G_0(\mathbf{x}, \mathbf{x}') = \delta(\mathbf{x} - \mathbf{x}')$$

Wave equation on $\mathcal{D}$ (scattering domain):

$$\partial_i \partial_i E(\mathbf{x}, \mathbf{x}') + \kappa^2 E(\mathbf{x}, \mathbf{x}') = J(\mathbf{x}, \mathbf{x}')$$

Operator notation

$$\mathbf{L}_0 \mathbf{G}_0 = \mathbf{I} \quad (1)$$

$$\mathbf{L} \mathbf{E} = \mathbf{J} \quad (2)$$

Scattering potential

$$\mathbf{L} - \mathbf{L}_0 = \mathbf{V} = \kappa^2 - \kappa_0^2$$

$$\Rightarrow \mathbf{L} = \mathbf{L}_0 + \mathbf{V}$$

Left-multiply (1) by $\mathbf{E}$ and (2) by $\mathbf{G}_0$ and subtract

$$\mathbf{G}_0 \mathbf{L} \mathbf{E} - \mathbf{E} \mathbf{L}_0 \mathbf{G}_0 = \mathbf{E} - \mathbf{G}_0 \mathbf{J}$$

$$\mathbf{E} = \mathbf{G}_0 \mathbf{J} + \mathbf{G}_0 (\mathbf{L} - \mathbf{L}_0) \mathbf{E}$$

Lippmann-Schwinger equation:

$$\mathbf{E} = \mathbf{G}_0 \mathbf{J} + \mathbf{G}_0 \mathbf{V} \mathbf{E} = \mathbf{E}_0 + \mathbf{G}_0 \mathbf{V} \mathbf{E}$$

Integral form

$$E(\mathbf{x}, \mathbf{x}_s) = E_0(\mathbf{x}, \mathbf{x}_s) + \int d^3 x' G_0(\mathbf{x}, \mathbf{x}') V(\mathbf{x}') E(\mathbf{x}', \mathbf{x}_s)$$

- So far this applies to both seismic, EM, ...

Consider EM, isotropic case:

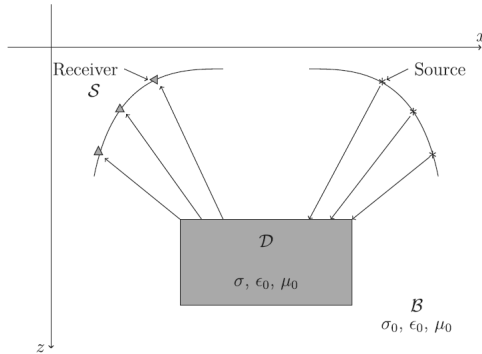
$$\kappa^2 = i\omega\mu_0\sigma$$

$$\kappa_0^2 = i\omega\mu_0\sigma_0$$

$$V = i\omega\mu_0(\sigma - \sigma_0) = i\omega\mu_0\sigma_0\chi = \kappa_0^2\chi$$

$$\chi = \frac{\Delta\sigma}{\sigma_0}$$

$$\Rightarrow \sigma = \sigma_0(1 + \chi)$$



$$\chi = 0 \text{ in } \mathcal{B}$$

$$\chi \neq 0 \text{ in } \mathcal{D}$$

Lippmann-Schwinger on $\mathcal{D}$

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{G}_0 \chi \mathbf{E}$$

Introduce the contrast sources

$$\mathbf{w} = \chi \mathbf{E}$$

Then

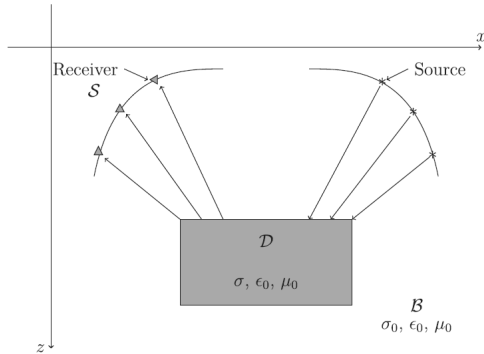
$$\mathbf{E} = \mathbf{E}_0 + \mathbf{G}_0 \mathbf{w}$$

Multiply with χ

$$\chi \mathbf{E} = \chi \mathbf{E}_0 + \chi \mathbf{G}_0 \mathbf{w} = \mathbf{w}$$

Written out; integral form:

$$w(\mathbf{x}) = \chi(\mathbf{x}) E_0(\mathbf{x}) + \chi(\mathbf{x}) \int_{\mathcal{D}} d^3 x' G_0^D(\mathbf{x}, \mathbf{x}') w(\mathbf{x}')$$



Data equation on $\mathcal{S}$ (recording surface):

$$\mathbf{F} = \mathbf{G}_0^S \chi \mathbf{E} = \mathbf{G}_0^S \mathbf{w}$$

Written out on integral form:

$$F(\mathbf{x}_S) = \int G_0(\mathbf{x}_S, \mathbf{x}') w(\mathbf{x}') d^3 x' \quad , \mathbf{x}_S \in \mathcal{S}$$

$F(\mathbf{x}_S)$ is the scattered-field data equation

CSI objective function:

$$\begin{aligned}\Phi = & \alpha_1 \|\Xi(\mathbf{F} - \mathbf{G}_0^S \mathbf{w})\|_S^2 \\ & + \alpha_2 \|\chi \mathbf{E}_0 + \chi \mathbf{G}_0 \mathbf{w} - \mathbf{w}\|_D^2 \\ & + \beta C\end{aligned}$$

— Data term

— Lippmann-Schwinger term

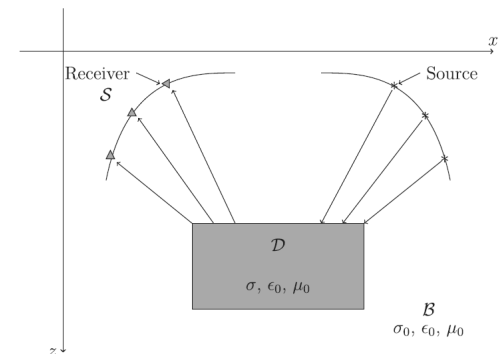
— Regularization

L_2 -norms:

$$\|\mathbf{f}\|_A^2 = \int_A |f(\mathbf{x})|^2 d^n x$$

Term 1: $\|\cdots\|_S^2 = \int_S |\cdots|^2 d^2 S$ (recording surface)

Term 2: $\|\cdots\|_D^2 = \int_D |\cdots|^2 d^3 x$ (domain $\mathcal{D}$)



Bayesian-style:

$$\begin{aligned}\Phi = & \alpha_1 [\mathbf{F} - \mathbf{G}_0^S \mathbf{w}]^H \boldsymbol{\Sigma}_F^{-1} [\mathbf{F} - \mathbf{G}_0^S \mathbf{w}] \\ & + \alpha_2 [\chi \mathbf{E} - \mathbf{w}]^H \boldsymbol{\Sigma}_w^{-1} [\chi \mathbf{E} - \mathbf{w}] \\ & + \beta \chi^H \boldsymbol{\Sigma}_\chi^{-1} \chi\end{aligned}$$

Data covariance:

$$\boldsymbol{\Sigma}_F^{-1} = \boldsymbol{\Xi}^T \boldsymbol{\Xi}$$

Smoothing:

$$\boldsymbol{\Sigma}_\chi^{-1} = \mathbf{D}^T \mathbf{D}$$

CSI - stepwise inversion approach

Step 1: Minimize wrt $\mathbf{w}$:

$$\begin{aligned}\Phi_1 = & \alpha_1 [\mathbf{F} - \mathbf{G}_0^S \mathbf{w}]^H \boldsymbol{\Sigma}_F^{-1} [\mathbf{F} - \mathbf{G}_0^S \mathbf{w}] \\ & + \alpha_2 [\chi \mathbf{E} - \mathbf{w}]^H \boldsymbol{\Sigma}_w^{-1} [\chi \mathbf{E} - \mathbf{w}]\end{aligned}$$

Initials:

$$\begin{aligned}\chi &= \chi_0 \\ \mathbf{E} &= \mathbf{E}_0 + \mathbf{G}_0^D \chi_0 \mathbf{E}_0\end{aligned}$$

Step 2: Compute (after update of $\mathbf{w}$):

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{G}_0^D \mathbf{w}$$

Step 3: Minimize wrt χ :

$$\Phi_2 = \alpha_2 [\chi \mathbf{E} - \mathbf{w}]^H \Sigma_w^{-1} [\chi \mathbf{E} - \mathbf{w}] + \beta \chi^T \Sigma_\chi^{-1} \chi$$

Note: this is a trivial inversion if $\beta = 0$:

$$\chi(\mathbf{x}) = \frac{w(\mathbf{x}) E^*(\mathbf{x})}{|E(\mathbf{x})|^2}$$

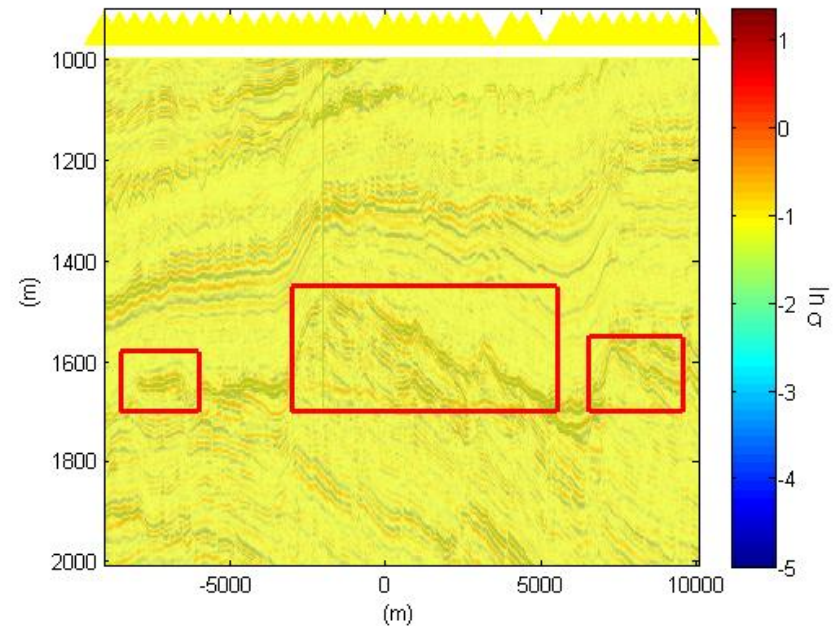
if $\Sigma_\chi^{-1} = \mathbf{D}^T \mathbf{D}$ we get a linear system of equations for χ

Example:

Troll CSEM inversion

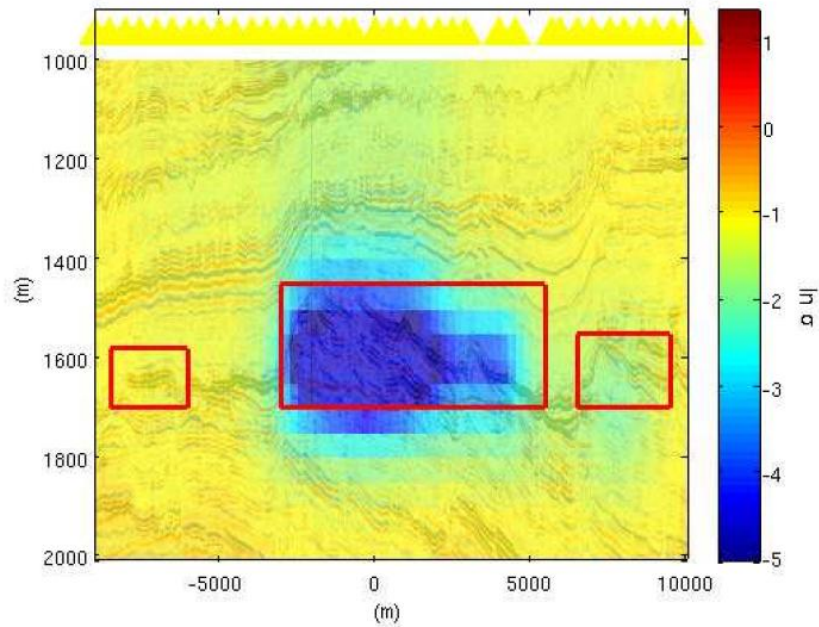
- Single line data from Troll Field
- 37 receivers, inline electric and crossline magnetic field
- 0.25Hz, 0.75Hz
- $\mathcal{D} : 22\text{km} \times 2\text{km} \times 1\text{km}$
- $\Delta x = 250\text{m}, \Delta y = 250\text{m}, \Delta z = 50\text{m}$

Wiik et al, (2011)

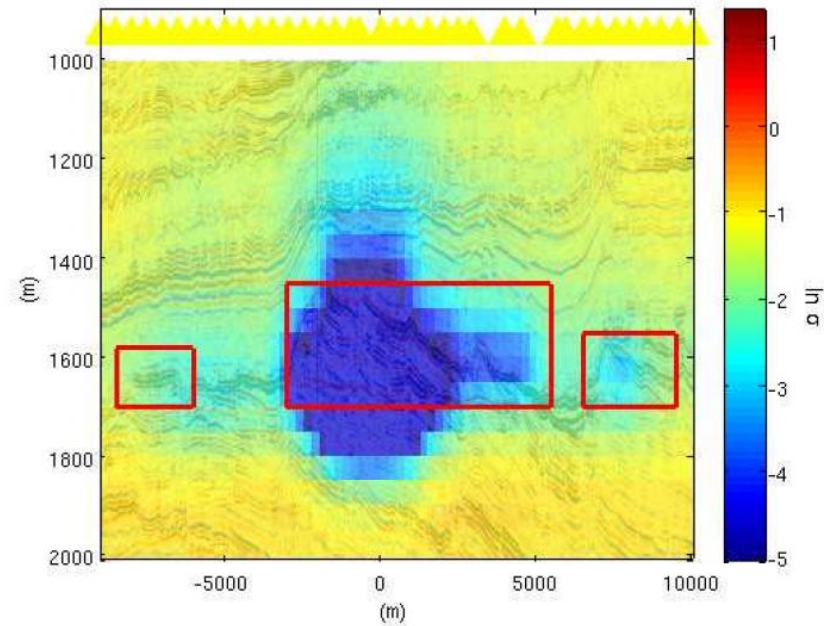


Background conductivity

CSI

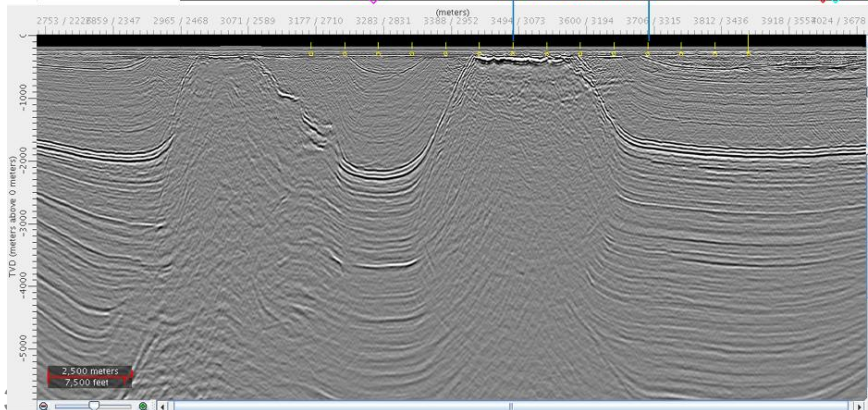
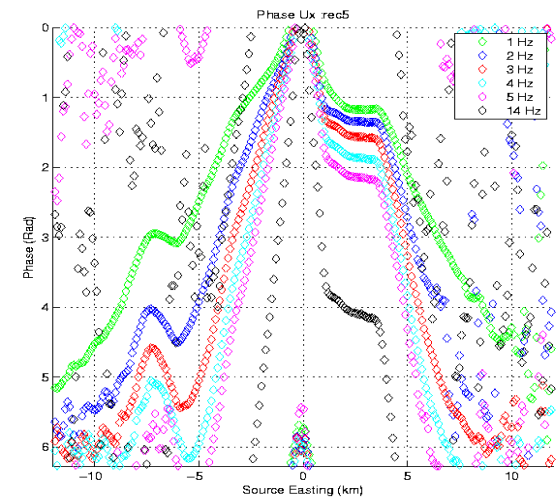
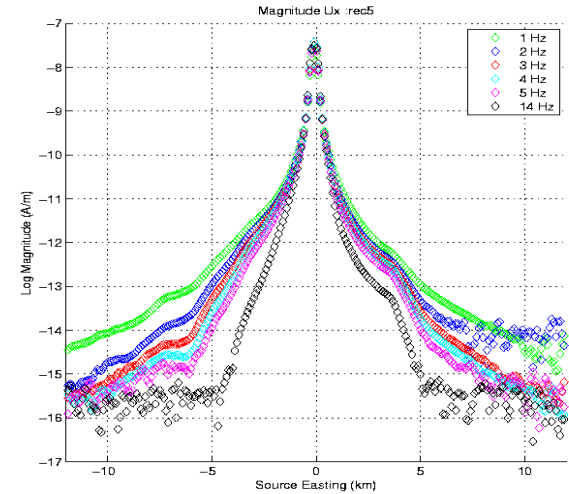
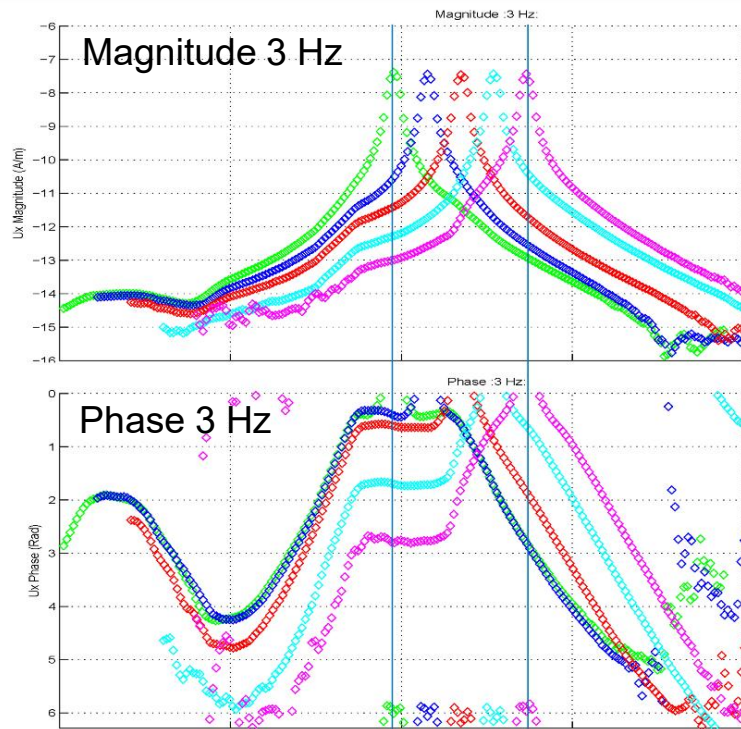


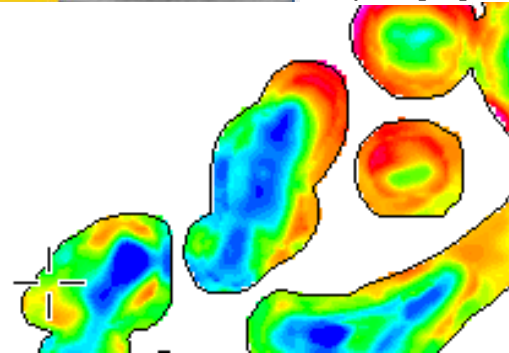
Isotropic



TIV, σ_V

Uranus CSEM survey

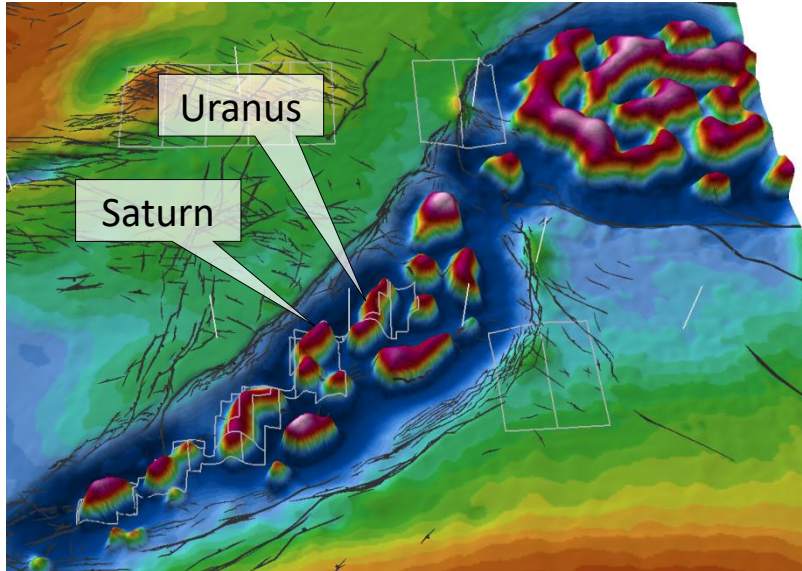




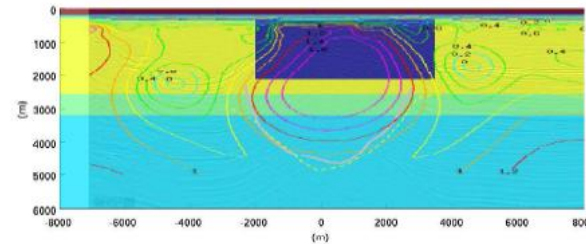
Hokstad et al. (2011), Joint Imaging of Geophysical Data:
Case History from the Nordkapp Basin, Barents Sea

Example:

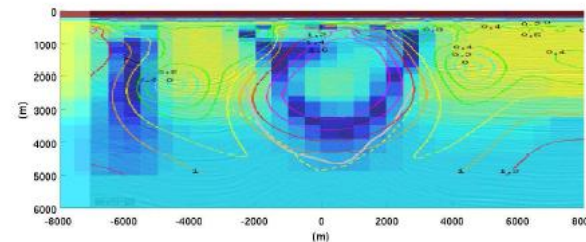
Nordkapp Basin joint CSEM and MT inversion



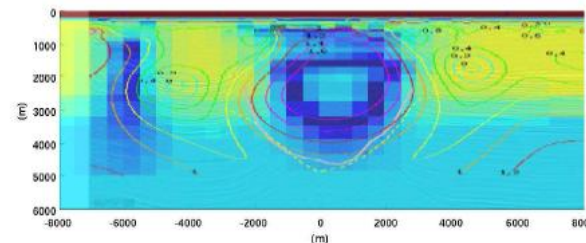
Wiik et al, (2013)



Initial model



MT inversion
(E and H fields)



MT+CSEM inversion

CSI summary

- Non-linear inversion by iteration of two linear steps
- CSI is applicable to any problem that has a Lippmann-Schwinger equation

Published applications:

- Medical acoustics (tumor imaging)
- GPR (waste management)
- CSEM and MT inversion (Wiik et al., 2011, 2013)